

PAULA FUDOLIG

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RESEARCH INTERESTS

Multi-wavelength studies of active galactic nuclei and black hole – galaxy co-evolution.

EDUCATION

George Mason University

Doctor of Philosophy in Physics

Aug. 2023 – Present

Fairfax, VA

Boston University

Bachelor of Arts in Astronomy and Physics

Aug. 2019 – May 2023

Boston, MA

PROJECTS

Radio Emission in Radio-Quiet Quasars

National Radio Astronomy Observatory

May 2025 – Present

Fairfax, VA

- **Performed** data calibration, imaging, and analysis of a sample of optically-selected quasars at 6 GHz to trace the origin of the observed radio-quiet emission.
- **Reduced** VLBA observations using the National Radio Astronomy Observatory's **CASA** and **AIPS** softwares.

Varstrometry for Dual AGN using Radio Interferometry

Computational Physics, Inc.

June 2023 – Present

Springfield, VA

- **Performed** data calibration, imaging, and analysis of a sample of quasars at 2GHz/8GHz to characterize source morphology on sub-milliarcsecond scales and investigate varstrometry+radio interferometry as a viable systematic detection method for the identification of multi-AGN.
- **Reduced** VLBA observations using the National Radio Astronomy Observatory's **CASA** and **AIPS** softwares, and **AOFlagger**.

Large VLBA BEAM-ME

Boston University Institute of Astrophysical Research

Jan. 2020 – May 2023

Boston, MA

- **Monitored** changing brightness and polarizations of a sample of blazar jets and their shockwaves at sub-milliarcsecond scales using circular Gaussian brightness distributions to model jet activity over time.
- **Determined** and compared significant blazar features to investigate correlations between jet activity and gamma-ray events.
- **Reduced** VLBA data using Caltech's **Difmap** program.

Patatrack Pixel Tracking

European Council for Nuclear Research (CERN)

Jan. 2022 – Aug. 2022

Geneva, Switzerland

- **Tested** Patatrack's 4 portability libraries of track reconstruction algorithms across 3 GPU devices on a varying amount of threads to evaluate performance.
- **Identified** performance bottlenecks of the algorithms for each architecture and implemented solutions.
- **Wrote** single-source code to run on all provided architectures to avoid unnecessary maintenance on various versions of the same algorithm.
- **Developed** **CUDA** skills in **C++** and **Python**, utilized **Git** in team setting.

SOFIA 3-Micron Dust Emission Studies

NASA Goddard Space Flight Center

June 2020 – Aug. 2020

Greenbelt, MD

- **Fitted** 3-5 micron wavelength emission spectra from polycyclic aromatic hydrocarbons surrounding a sample of planetary nebulae to study the relationship between hydrocarbon features and planetary nebulae ages.
- **Compared** Gaussian and Lorentzian distributions of data to establish a standard fitting model.
- **Programmed** in **Python**, utilizing astronomical data from ground and airborne observations.

PROPOSALS & GRANTS

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| Tracing the Origin of AGN Emission in Variable Radio-Quiet Quasars | Semester 2026A |
| <ul style="list-style-type: none">• Principal investigator. | |
| VLBA Characterization of Astrometrically-Variable Quasars | Semester 2024B |
| <ul style="list-style-type: none">• Principal investigator. | |
| Undergraduate Research Opportunities Program Grant | June 2022 – Aug. 2022 |
| <ul style="list-style-type: none">• Over \$4000 in funding to extend internship at the European Council for Nuclear Research (CERN). | |

PUBLICATIONS

- Fudolig, P.**, Kimball, & Momjian. Constraining the Origins of Radio Emission in Radio-quiet QSOs using the Very Long Baseline Array, in prep.
- Fudolig, P.**, et al. Varstrometry for Dual AGN using Radio Interferometry: VaDAR with the VLBA II, in prep.
- Schwartzman, E., **Fudolig, P.**, et al. Varstrometry for Dual AGN using Radio Interferometry: VaDAR with the VLBA, 2025, ApJ, 987, 200

CONFERENCES, COLLOQUIA, & PRESENTATIONS

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| DC Astrophysics Graduate Student Conference | Aug. 14-15, 2025 |
| <i>Talk title: Tracing the Origins of Radio Emission in Radio-Quiet Quasars</i> | <i>Washington, D.C.</i> |
| 245th American Astronomical Society Meeting | Jan. 12-16, 2025 |
| <i>Talk title: A VLBA View of VaDAR</i> | <i>National Harbor, MD</i> |
| 11th Annual Mid-Atlantic Radio-Loud AGN Meeting | November 8, 2024 |
| <i>Talk title: A VLBA View of VaDAR</i> | <i>Baltimore, MD</i> |
| DC Astrophysics Graduate Student Conference | Aug. 15-16, 2024 |
| <i>Talk title: A VLBA View of VaDAR</i> | <i>Washington, D.C.</i> |
| 10th Annual Mid-Atlantic Radio-Loud AGN Meeting | October 12, 2023 |
| <i>Talk title: VaDAR: Varstrometry for Dual AGN using Radio Interferometry</i> | <i>Fairfax, VA</i> |
| Boston University Undergraduate Research Symposium | Oct. 21, 2022 |
| <i>Poster title: Pixel-Tracking in CERN's Compact Muon Solenoid</i> | <i>Boston, MA</i> |
| NASA Goddard Summer Poster Session | August 4, 2020 |
| <i>Presentation title: Comparing Fits of PAH Emission</i> | <i>Greenbelt, MD</i> |

SERVICE

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| SPECTRUM Leader | March 2024 – Present |
| <i>Organization for enhancement of historically minoritized groups in STEM</i> | <i>Fairfax, VA</i> |
| <ul style="list-style-type: none">• Created and implemented a series of lunch workshops promoting the professional development of graduate and undergraduate students within the College of Science.• Created and implemented a mentorship program for incoming graduate students, including developing mentor/mentee applications, matching mentoring pairs, advertising the opportunity, and organizing related social events. Resulted in a 100% retention rate upon its first implementation in Spring 2024. | |
| Student Advisor | August 2025 – Present |
| <i>GMU College of Science Dean's Student Advisory Board</i> | <i>Fairfax, VA</i> |
| <ul style="list-style-type: none">• Shared insights and feedback on the student experience, representing graduate student needs and interests.• Collaborated with the Dean's office on college-level initiatives and improvements.• Fostered a stronger sense of community across programs and disciplines within the College of Science. | |
| Physics Department Representative | August 2024 – May 2025 |
| <i>George Mason's Graduate and Professional Student Association</i> | <i>Fairfax, VA</i> |
| <ul style="list-style-type: none">• Advocated for fellow Physics graduate students on topics including graduate tuition and fees, fellowship opportunities, student life, and campus events. | |

- **Collaborated** with other graduate student representatives from all departments on University-level initiatives.
- **Communicated** important information on departmental changes and solicited feedback for upcoming votes.

Peer Mentor

Sep. 2021 – May 2023

Boston University's PeerRs for Incoming Student Mentorship program

Boston, MA

- **Guided** first-year students by providing departmental and campus resources, creating four-year plans, sharing personal experiences, and providing support throughout the academic year.

Founder of the BU Society of Women in Space Exploration

Aug. 2020 – May 2023

Organization supporting the inclusion of women and minorities interested in space-related fields

Boston, MA

- As Founder: **authored** organization's constitution and bylaws, assembled an executive board, recruited a faculty advisor, and advertised on campus to build membership.
- As President: **led** biweekly general member meetings, involving devising group activities, planning on-campus events, and networking in the Boston area for collaborations. Also led executive board meetings, delegating tasks and overseeing management of responsibilities.

RELATED EXPERIENCE

Graduate Teaching Assistant

Aug. 2023 – Present

- Led undergraduate labs in Physics I/II and Astronomy I at George Mason University.

NRAO Socorro Lunch Talks

January 21, 2026

- Presented "A Tale of Two Variabilities: VaDAR and the Brightening J0822+4553".

Astronomy on Tap Speaker

November 18, 2025

- Presented "Hunting for Dual AGN with VaDAR" to the general public.

Observation Time on Perkins Telescope

March 8-10, 2023

- Three nights operating Boston University's Perkins Telescope at Anderson Mesa Observatory in Flagstaff, AZ.

SCIENTIFIC & TECHNICAL SKILLS

World Languages: English (native), Spanish (advanced), French (intermediate), Tagalog (basic)

Programming: Python, C++, IDL, CUDA, ROOT, Git, LaTeX, Ubuntu, Visual Studio Code, Jupyter, Mathematica, Arduino, Bash/Shell, Linux, Mac OS, Windows OS

Astronomical: Maxim DL, SAOImageDS9, Caltech Difmap, NRAO CASA, NRAO AIPS, PyBDSF, TopCat